Chirality kernels and the smallest regular cover

Human Compression project

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Abstract

We recover all proved conclusions of Cunningham's eight-page author manuscript of 25 July 2023. An intersection lemma reduces polytopality of the mirror mix to the intersection of two embedded chirality groups. We verify the local mirror identifications and the regular-end induction before making that reduction. The applications include chiral square torus maps, all five specified atomic tight families, and an exact finite computation producing two distinct minimal regular polytope covers. The older classification needed for the five families is retained explicitly. Later results about the source's questions are recorded separately and are not substitutes for the source-specific proofs.

1 Compatible sections and their mirror mixes

Let P be a chiral abstract n-polytope with a chosen base flag. We use the common paper's string rotation notation

$$\Gamma = U_n/N = \langle \sigma_1, \dots, \sigma_{n-1} \rangle, \qquad \Lambda = \Gamma \diamond \overline{\Gamma} \cong U_n/(N \cap \overline{N}).$$

The symbol $\diamond$ means the subgroup of the direct product generated by the corresponding pairs of marked generators. It does not by itself assert polytopality. The common invariant-kernel theorem proves that Λ is the rotation group of the smallest regular maniplex covering P. We denote this object by R. Its polytopality is the question to be tested.

Let K, L, M be the facet, vertex-figure and medial section through the base flag. All their rotation groups are embedded in the same ambient Γ. Put

$$A = \Lambda_{n-1}, \qquad B = \Lambda_0, \qquad C = \Lambda_{0,n-1}, \qquad \pi : \Lambda \longrightarrow \overline{\Gamma}.$$

The common local-mirror lemma gives

$$A \cap \ker \pi = X(K) \times 1, \quad B \cap \ker \pi = X(L) \times 1, \quad C \cap \ker \pi = X(M) \times 1. \tag{1}$$

The conjugation of the second coordinate in the vertex and medial cases is important: the global mirror is conjugation by r_0, whereas the intrinsic mirror of those sections is conjugation by r_1. Their relation $r_0 = s_1 r_1$ supplies exactly that conjugation. An abstract isomorphism between two chirality groups would not establish (1).

We retain the standard string C^+ correspondence, with its recursive end-intersection criterion, as stated fully in the common paper. We also retain the Schulte–Weiss section theorem: facets of facets and vertex-figures of vertex-figures of a chiral polytope are directly regular. All sections of a directly regular polytope are directly regular. The next lemma proves the needed cover statement instead of retaining Cunningham's Proposition 2.2 as an input.

Lemma 1.1 (A regular end)**.** *If a chiral or directly regular polytope has directly regular facets or directly regular vertex-figures, its smallest regular cover is a polytope.*

Proof. Induct on rank. Polygons have cyclic rotation group and are directly regular; the rank-one case is also immediate. A directly regular polytope is already its own smallest regular cover. Suppose now the facets K are directly regular. Their mirror mixes are graphs of automorphisms, so A is a marked copy of the facet rotation group, is polytopal, and satisfies $A \cap \ker \pi = 1$. The facets of a vertex-figure L are vertex-figures of the facets K, and hence are directly regular. The induction hypothesis therefore makes the mirror mix B polytopal as well.

The second-coordinate marked group is a copy of the original mirrored polytope, so its intersection property gives

$$\pi(A) \cap \pi(B) = \pi(C).$$

The kernel condition in the common intersection-transport theorem holds because $A \cap \ker \pi = 1$. Thus $A \cap B = C$. Both ends are polytopal, so the recursive string C^+ criterion makes Λ polytopal. If the vertex-figures are directly regular instead, use the dual argument. This proves the induction without assuming the result to be established. □

For arbitrary chiral P, the section theorem now ensures that both A and B are polytopal. A chiral facet has directly regular facets, so Lemma 1.1 applies to its mirror mix; the dual statement handles a chiral vertex-figure, and a directly regular end needs no induction. Consequently R is a polytope exactly when $A \cap B = C$. This is the local prerequisite behind source Proposition 2.3.

Theorem 1.2 (Cunningham's criterion)**.** *For compatibly marked sections K, L, M of a chiral polytope P,*

$$R \text{ is a polytope} \quad \Longleftrightarrow \quad X(K) \cap X(L) \leq X(M).$$

Proof. The image equality $\pi(A) \cap \pi(B) = \pi(C)$ holds as above. The common intersection-transport theorem turns $A \cap B = C$ into $A \cap B \cap \ker \pi \leq C$. Formula (1) identifies this condition exactly with the displayed inclusion. The preceding paragraph supplies the polytopality of both ends, so the equality is equivalent to polytopality of R. □

2 All general consequences in the source

Corollary 2.1. *The following statements hold with all subgroups embedded using the chosen base flag.*

(a) If either end section is directly regular, R is polytopal.

(b) If M is directly regular, R is polytopal exactly when $X(K) \cap X(L) = 1$. Equivalently, this intersection meets $\Gamma_{0,n-1}$ trivially.

(c) In rank four, R is polytopal exactly when $X(K) \cap X(L)$ contains no nonidentity power of σ_2.

(d) If K and L are totally chiral, then R is polytopal exactly when M is totally chiral.

(e) If a chiral polytope K satisfies $X(K) \cap \Gamma_0(K) = 1$, then every chiral polytope having K as facet has a polytopal smallest regular cover.

Proof. A directly regular section has trivial chirality group, proving (a). In (b), $X(M) = 1$; also

$$X(K) \cap X(L) \leq \Gamma_{n-1} \cap \Gamma_0 = \Gamma_{0,n-1},$$

so taking a further intersection with the medial group does not change the condition. For (c), the medial section is a polygon, its rotation group is $\langle \sigma_2 \rangle$, and its chirality group is trivial. Every nontrivial element in the displayed intersection is therefore a nonidentity power of σ_2. The word "nonidentity" is necessary: the identity is in every subgroup.

For (d), total chirality means $X(K) = \Gamma_{n-1}$ and $X(L) = \Gamma_0$. Their intersection is the full medial group, so the criterion holds exactly when $X(M) = \Gamma_{0,n-1}$. Finally, in (e) the common medial subgroup is the subgroup $\Gamma_0(K)$ inside the facet. Hence $X(K) \cap X(L)$ is contained in the assumed trivial intersection, proving the criterion for every such extension. This proves no existence statement for extensions of a prescribed rank. □

These are respectively source Remark 3.2, Corollaries 3.3–3.4, Proposition 3.6 and Corollary 3.7. The criterion itself is source Theorem 3.1. The first-coordinate identity for chirality in the common paper recovers source Proposition 2.4; the argument above also recovers Propositions 2.2–2.3.

3 The two infinite-family applications

3.1 Chiral square torus maps

For the chiral maps $\{4,4\}_{(b,c)}$ in the usual square-torus parameter convention, the source assumes $bc(b-c) \neq 0$. We retain the standard identification with a quotient of the square tessellation. In that identification the oriented marked group is

$$\big(\mathbb{Z}[i]/(b+ic)\big) \rtimes \langle i\rangle, \qquad u : z \mapsto iz, \qquad v : z \mapsto iz+1.$$

The affine notation describes elements of the abstract semidirect product; even a degenerate small quotient is not being declared an abstract polytope merely from its permutation action. The product vu^{-1} is translation by 1, and conjugation by u gives translation by i, so the indicated elements generate all translations and the quarter-turn. Their products satisfy $(uv)^2 = 1$. Also $u^4 = v^4 = 1$, since the successive translation terms for v^4 sum to $1+i-1-i = 0$. The linear-part homomorphism ℓ onto C_4 sends both u and v to i, so both generators have exact order four.

On the universal marked group the mirror replaces (u, v) by (u^{-1}, u^2v). Both new linear parts are i^{-1}. Thus $\ell\mu$ is inversion composed with ℓ. The relation kernel and its mirror both lie in the kernel of this linear-part map. Their product does too, and consequently

$$X(K) \leq \ker \ell.$$

But ℓ is injective on $\langle v\rangle$, since v and $\ell(v)$ both have order four. It follows that $X(K) \cap \langle v\rangle = 1$. Corollary 2.1(e) proves source Example 3.8 for every such chiral facet. No noncanonical or negative-parameter extension of the source's chiral-map assertion is implicit. The quotient C_4 used by the proof is a regular marked group; we do not need the one-cell square quotient $\{4,4\}_{(1,0)}$ to be polytopal.

3.2 The five atomic tight families

The older input here is Cunningham–Pellicer, *Tight chiral polytopes*, Journal of Algebraic Combinatorics 54 (2021), 837–878. We retain the existence and atomicity of its five Table 4 families having both ends chiral, together with its published Proposition 11. The acquired arXiv version 2009.04566v1 labels the relevant result Proposition 4.2, on printed page 20. The earlier review read that proposition, proof, neighboring embedded-group arguments and the five table rows; it did not independently reprove the full classification.

For those atomic polytopes the retained result gives

$$X(P) = \langle \sigma_2^{q'} \rangle, \qquad q/q' \text{ prime},$$

where q is the middle generator order. The local chirality groups are nontrivial because both ends are chiral. Each embeds in the global $X(P)$: its first-coordinate kernel is obtained by restricting the global mirror mix to an end subgroup. A nontrivial subgroup of the cyclic prime-order $X(P)$ must equal it. Thus

$$X(K) = X(L) = X(P)$$

as subgroups of the same Γ, not merely as abstract isomorphism types.

Type	Parameters in the retained family	Common nonidentity element
$\{2m, m^\alpha, 2m\}$	m odd prime, $\alpha \geq 2$, $1 \leq k \leq m-1$	$\sigma_2^{m^{\alpha-1}}$, order m
$\{8, 2^\beta, 8\}$	$\beta \geq 5$, $\epsilon_1, \epsilon_2 \in \{\pm 1\}$	$\sigma_2^{2^{\beta-1}}$, order 2
$\{2^{\beta-1}, 2^\beta, 2^{\beta-1}\}$	same range	same element
$\{8, 2^\beta, 2^{\beta-1}\}$	same range	same element
$\{2^{\beta-1}, 2^\beta, 8\}$	same range	same element

The parameters specify the classified marked groups, not every polytope with the same Schläfli type. The indicated elements are nonidentity because the middle generator has exact order m^α or 2^β. Corollary 2.1(c) therefore shows that the smallest regular cover of every polytope in these five families is not polytopal. This is the full all-parameter deduction in source Example 3.5. It uses the older classification as stated; finite sampling cannot replace it.

4 Two distinct finite minimal covers

Use a, b, c for rotation generators, with common relations

$$(ab)^2 = (bc)^2 = (abc)^2 = 1. \tag{2}$$

Let P have, in addition, the literal relations

$$a^6 = b^9 = c^6 = 1, \qquad ba^2 = a^2b^4, \qquad c^2b = b^4c^2. \tag{3}$$

Let T have (2) and

$$a^6 = b^9 = c^2 = 1, \qquad (ab^{-2})^2 = 1, \qquad ab^2a(ab^{-1}a)^2 = 1, \tag{4}$$

and let T^δ have (2) and

$$a^2 = b^9 = c^6 = 1, \qquad (c^{-1}b^2)^2 = 1, \qquad bc^2b(c^{-1}bc^{-1})^2 = 1. \tag{5}$$

These are the three presentations printed on page 7 of the checked author edition. No extra relation is added to make an order match. We set

$$R = P \diamond \overline{P}, \qquad Q = R \diamond T, \qquad Q' = R \diamond T^\delta,$$

where the mirror marking of P is (a^{-1}, a^2b, c).

The complete required program `chiral_cover.g` starts with the three finitely presented groups, obtains exact GAP isomorphisms to finite permutation groups, and generates the mixes inside direct products from the literal paired generators. It verifies the following table. Every order in the table is a rotation-group order; "mirror" means that the displayed mirror assignment extends to an automorphism.

Group	Order	Orders of a, b, c	End orders	End intersection	Mirror
P	324	6, 9, 6	54, 54	9	no
T	324	6, 9, 2	162, 18	9	yes
T^δ	324	2, 9, 6	18, 162	9	yes
R	972	6, 9, 6	162, 162	27	yes
Q	2916	6, 9, 6	162, 162	9	yes
Q'	2916	6, 9, 6	162, 162	9	yes

For every row the program also checks the rotation relations, $|\langle b\rangle| = 9$, and

$$\langle a\rangle \cap \langle b\rangle = 1, \qquad \langle b\rangle \cap \langle c\rangle = 1.$$

For the five successful rows it checks the actual subgroup equality $\langle a, b\rangle \cap \langle b, c\rangle = \langle b\rangle$; for R it checks failure. These equalities are not inferred from similarly sized but unidentified groups. The table therefore proves, using the retained recursive correspondence, that P is chiral, T, T^δ, Q, Q' are directly regular polytopes, and R is not a polytope. For P, the specified mirror assignment fails to define a homomorphism. For every positive row the assignment is checked to be bijective.

The program computes the first-coordinate images of the second-projection kernels in the two ends of R. Both are exactly $\langle b^3\rangle \leq \Gamma(P)$, of order three. Thus the finite example exhibits directly the obstruction in Theorem 1.2: its medial polygon has trivial chirality, whereas b^3 lies nontrivially in both end chirality groups.

It also verifies a marked isomorphism from the facet of T to the facet of R, and the dual isomorphism

$$(a, b, c)_T \longmapsto (c^{-1}, b^{-1}, a^{-1})_{T^\delta}.$$

The facet subgroup of T has index two, so the directly regular polytope T has exactly two facets. Under the standard interpretation of the trivial extension used by the source, this is the trivial extension of that facet. The dual map proves the claimed identity of T^δ, not just an agreement of orders.

The marked epimorphisms

$$R \to P, \qquad Q \to R, \qquad Q' \to R$$

are verified and each has kernel order three. The common invariant-kernel theorem proves smallestness of R; the common prime-degree lemma then proves minimality of Q and Q' as regular polytope covers of P. Doubling rotation orders gives the source's respective flag counts 648, 1944 and 5832.

To distinguish the two minimal covers, the program supplies the word

$$w = (bc^{-1})^2 ac^2 a^{-1}. \tag{6}$$

Its order is 1 in Q and 3 in Q'. Hence no marked isomorphism exists. This also excludes an unbased rank-preserving polytope isomorphism. Such an isomorphism commutes with every flag-adjacency operation. Composing it with an automorphism of the regular target carries the image of a chosen base flag back to that target's chosen base flag; it would then identify all distinguished rotation generators. It would preserve the value and order of (6), a contradiction. No assertion about dual isomorphisms follows from this word test.

The calculation retains GAP's exact algorithms for finitely presented groups, permutation isomorphisms, subgroup intersections and homomorphisms. It uses neither RAMP nor a group catalogue. The source used RAMP for its example; the replacement program is the previously reviewed bare-presentation computation, copied in full. Its inputs and logical outputs are all specified above. The uniform criterion and the two infinite-family arguments do not depend on the finite table.

5 Attribution, editions and the separate questions

The proved source conclusions covered here are Propositions 2.2–2.4, Theorem 3.1, Remark 3.2, Corollaries 3.3–3.4, Example 3.5, Proposition 3.6, Corollary 3.7, Example 3.8, the uniqueness implication at the beginning of Section 4 and all claims in its explicit finite example. The exact source is Gabe Cunningham, *Chiral polytopes whose smallest regular cover is a polytope*, the complete eight-page author manuscript headed 25 July 2023. Publication metadata is Journal of Combinatorial Theory, Series A 204 (2024), 105839. The full publisher PDF was not acquired, so this is not a certification

of identical wording in the version of record. The earlier publisher metadata notes a correction concerning Corollary 3.3; our proof binds the precise embedded-subgroup statement in the checked author edition.

The common group calculations are elementary reformulations of arguments already used in this literature and the accepted proof bank. The source's own propositions are not assumed to prove its main theorem. The string C^+ correspondence and depth-two section theorem remain retained geometric inputs; the square-tessellation model and the specified atomic-family classification remain additional retained inputs. The finite groups are established by the complete required GAP program. The stated Lean certificates verify their exact generic interfaces, not these source-specific geometric or computational instantiations.

The source also asks whether nonpolytopality of the smallest regular cover always forces multiple minimal regular polytope covers, and conjectures the analogous phenomenon for all five tight families in Example 3.5. These are not proved by the arguments of this application. Later admitted project results concerning them have their own proofs, reviews and quantifiers: the general statement allowing arbitrary regular covers, finite rank-four results, and interval-extension results must not be conflated with a theorem about finite covers in every rank. Their status is recorded in the accompanying question map. Those later proofs are not used as unpriced premises here, and no new question credit is assigned to this canonical edition.

The statement requiring finite covers in every rank remains unresolved in the closed corpus. Results for arbitrary independent markings do not automatically apply to the monodromy groups of actual polytopes. This application preserves the source's complete proved conclusions without importing a stronger claim from that separate investigation.